

Problem

Evaluate these determinants:

$$(a) \begin{vmatrix} 10 + j6 & 2 - j3 \\ -5 & -1 + j \end{vmatrix}$$

$$(b) \begin{vmatrix} 20\angle -30^\circ & -4\angle -10^\circ \\ 16\angle 0^\circ & 3\angle 40^\circ \end{vmatrix}$$

$$(c) \begin{vmatrix} 1 - j & -j & 0 \\ j & 1 & -j \\ 1 & j & 1 + j \end{vmatrix}$$

Step-by-step solution

Step 1 of 3

(a)

Calculate the determinant of $\begin{vmatrix} 10 + j6 & 2 - j3 \\ -5 & -1 + j \end{vmatrix}$.

$$\begin{aligned} \begin{vmatrix} 10 + j6 & 2 - j3 \\ -5 & -1 + j \end{vmatrix} &= (10 + j6)(-1 + j) - (2 - j3)(-5) \\ &= (-10 + j10 - j6 - 6) - (-10 + j15) \\ &= -16 + j4 + 10 - j15 \\ &= -6 - j11 \end{aligned}$$

Therefore, the determinant of $\begin{vmatrix} 10 + j6 & 2 - j3 \\ -5 & -1 + j \end{vmatrix}$ is $\boxed{-6 - j11}$.

Step 2 of 3

(b)

Calculate the determinant of $\begin{vmatrix} 20\angle -30^\circ & -4\angle -10^\circ \\ 16\angle 0^\circ & 3\angle 40^\circ \end{vmatrix}$.

$$\begin{aligned} \begin{vmatrix} 20\angle -30^\circ & -4\angle -10^\circ \\ 16\angle 0^\circ & 3\angle 40^\circ \end{vmatrix} &= (20\angle -30^\circ)(3\angle 40^\circ) - (-4\angle -10^\circ)(16\angle 0^\circ) \\ &= 60\angle 10^\circ + 64\angle -10^\circ \\ &= 59.089 + j10.42 + 63.027 - j11.11 \\ &= 122.12 - j0.69 \end{aligned}$$

Therefore, the determinant of $\begin{vmatrix} 20\angle -30^\circ & -4\angle -10^\circ \\ 16\angle 0^\circ & 3\angle 40^\circ \end{vmatrix}$ is $\boxed{122.12 - j0.69}$.

Step 3 of 3

(c)

Calculate the determinant of $\begin{vmatrix} 1-j & -j & 0 \\ j & 1 & -j \\ 1 & j & 1+j \end{vmatrix}$.

$$\begin{aligned} \begin{vmatrix} 1-j & -j & 0 \\ j & 1 & -j \\ 1 & j & 1+j \end{vmatrix} &= (1-j) \begin{vmatrix} 1 & -j \\ j & 1+j \end{vmatrix} + j \begin{vmatrix} j & -j \\ 1 & 1+j \end{vmatrix} + 0 \begin{vmatrix} j & 1 \\ 1 & j \end{vmatrix} \\ &= (1-j)[(1+j) + j^2] + j[j(1+j) + j] \\ &= (1-j)[(1+j) - 1] + j[j - 1 + j] \\ &= (1-j)[j] + j[-1 + 2j] \\ &= j(1 - j - 1 + 2j) \\ &= j(j) \\ &= -1 \end{aligned}$$

Therefore, the determinant of $\begin{vmatrix} 1-j & -j & 0 \\ j & 1 & -j \\ 1 & j & 1+j \end{vmatrix}$ is $\boxed{-1}$.